ARRAY

Coding Questions

Pairs

Given an array containing N integers, and a number S denoting a target sum.

Find two distinct integers that can pair up to form target sum. Let us assume there will be only one such pair.

Input:
7
10 5 2 3 -6 9 11
4

Output:
10 -6

Count the Triplets

Given an array arr[ ] of N distinct integers. The task is to find the number of triplets (i, j, k), where i, j, k are indices and (1 <= i < j < k <= N), such that in the set $\{A_i, A_j, A_k\}$ at least one of the numbers can be written as the sum of other two.

Input:
5
1 2 3 4 5

Output:
4
(1, 2, 3), (1, 3, 4), (1, 4, 5), (2, 3, 5)

Subarray with given sum

Given an array arr[ ] of non-negative integers and an integer sum. Find the sub-array that adds to the given sum. Print the start and end index of the sub-array.

NOTE:
There may be more than one sub-array with sum as given sum , print the first such array.

Input:
1 4 20 3 10 5
33

Output:
2 4

Missing number in an array

Given an array nums containing N distinct numbers in the range [0, N], return the only number in the range that is missing from the array.

Input:
3
3 0 1

Output:
2

Pythagorean Triplet

Given an array of N integers, write a function that returns true if there is a triplet (a, b, c) that satisfies $a^2 + b^2 = c^2$

Input:
5
3 1 4 6 5

Output:
True

There is a Pythagorean triplet (3, 4, 5).

Mountain

Write a function that takes input an array of distinct integers, and returns the length of the highest mountain.

A mountain is defined as adjacent integers that are strictly increasing until they reach a peak, at which they become strictly decreasing.

At least 3 numbers are required to form a mountain.

Input:
16
5 6 1 2 3 4 5 4 3 2 0 1 2 3 -2 4

Output:
9

Chocolate Distribution Problem

Given an array of N integers where each value represents the number of chocolates in a packet. Each packet can have a variable number of chocolates. There are m students, the task is to distribute chocolate packets such that:

- **Each student gets one packet.**
- **The difference between the number of chocolates in the packet with maximum chocolates and the packet with minimum chocolates given to the students is minimum.**

Input:
8
3 4 1 9 56 7 9 12
5

Output:
6

*3, 4, 7, 9, and 9 is been distributed among the 5 children, so the difference between the number of chocolates in the packet with **maximum** chocolates and packet with **minimum** chocolates would be **9 - 3 = 6***

Rains

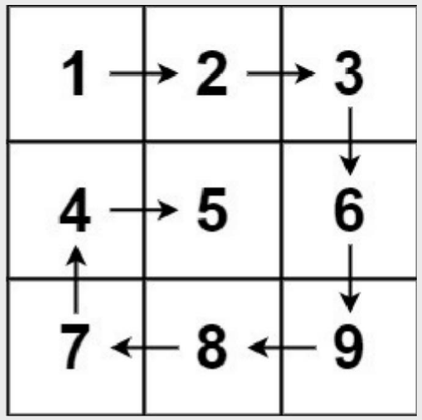
Given N non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.

Input:
12
0 1 0 2 1 0 1 3 2 1 2 1

Output:
6

Spirally traversing a matrix

Given an mXn matrix, return all elements of the matrix in spiral order.



Input:
3 3
1 2 3
4 5 6
7 8 9

Output:
1 2 3 6 9 8 7 4 5

Largest Number formed from an Array

Given a list of non-negative integers *nums* of size N, arrange them such that they form the largest number and return it.

Since the result may be very large, so you need to return a string instead of an integer.

Input:
5
3 30 34 5 9

Output:
“9534330”

K-th smallest element

Given an array of N numbers and a number K where K < N. Find the K-th smallest element in the given array. Given that all array elements are distinct.

Input:
5
12 3 5 7 19
2

Output:
5